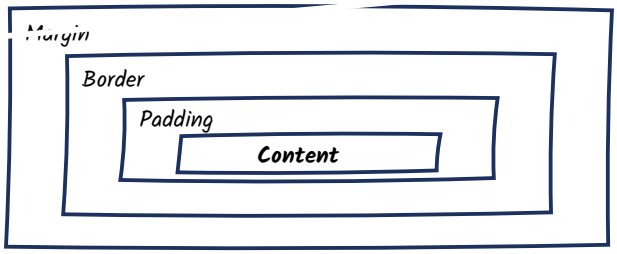
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Q1. What is CSS Box Model?

Ans: The CSS Box Model describes how every HTML element is treated as a box. It consists of four parts:

- **Content** – The actual text, image or other content of the element.
- **Padding** – Space between the content and the border.
- **Border** – A line surrounding the padding and content.
- **Margin** – Space outside the border, used to separate elements.



Q2. What is media query? Explain with a sample code.

Ans: A media query is a CSS technique used to apply different styles according to the screen size or device. It is mainly used to make a webpage responsive.

Example:

```
@media (max-width: 600px) {
    body {
        background-color: lightblue;
        font-size: 14px;
    }
}
```

Here, the styles inside the media query are applied when the screen width is 600px or less.

Q3. What are the features of Bootstrap?

Ans: Bootstrap is a popular CSS framework used for designing responsive and attractive web pages. Its main features are:

1. **Responsive design** : Pages work well on different screen sizes.
2. **Grid system** : Helps to arrange content in rows and columns.
3. **Pre-designed components** : Buttons, cards, forms, navbar, etc.
4. **Utility classes** : Ready-made classes for spacing, colors and alignment.
5. **Cross-browser support** : Works on modern web browsers.
6. **Easy to use** : Helps in faster web development.

Q4. Write an example of basic grid structure in Bootstrap?

Ans: Bootstrap grid divides the page into 12 columns. A basic grid structure can be written as:

```
<div class="container">
    <div class="row">
        <div class="col-md-6">Column 1</div>
```

```
        <div class="col-md-6">Column 2</div>
    </div>
</div>
```

Here, the row contains two columns and each column takes 6 out of 12 grid columns.

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Q5. What is CSS Selectors?

Ans: CSS selectors are used to select HTML elements on which CSS styles are applied. Common selectors are:

Selector	Meaning	Example
Element	Selects all elements of a type	`p { color: blue; }`
Class	Selects elements with a class	`.box { ... }`
ID	Selects one element by id	`#header { ... }`
Universal	Selects all elements	`* { margin: 0; }`
Group	Selects multiple elements	`h1, p { ... }`

Q6. What are CSS backgrounds, list the properties?

Ans: CSS background properties are used to set the background of an HTML element. Important properties are:

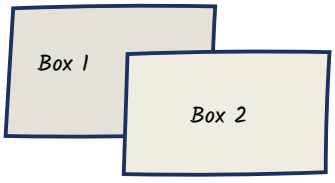
Property	Use
background-color	Sets the background color.
background-image	Sets an image as the background.
background-repeat	Controls whether the image repeats.
background-position	Sets the position of the image.
background-size	Controls the size of the background image.
background-attachment	Controls whether the background scrolls.
background	Shorthand for background properties.

Q7. How can we overlap elements in CSS?

Ans: Elements can be overlapped by using CSS position properties and the `z-index` property. An element can be positioned absolutely inside a relatively positioned parent.

```
.box1 { position: absolute; z-index: 1; }
.box2 { position: absolute; z-index: 2; }
```

The element with the higher `z-index` appears above the element with the lower `z-index`.



Q8. What are the different positioning properties in CSS?

Ans: The position property controls how an element is placed on a webpage. The main values are:

Position	Description
static	Default position of an element.
relative	Moves relative to its normal position.
absolute	Positioned relative to the nearest positioned ancestor.
fixed	Stays fixed relative to the browser window.
sticky	Acts relative until a scroll point, then behaves like fixed.

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Q9. What is ECMAScript?

Ans: ECMAScript is a standard specification for scripting languages. JavaScript is the most well-known implementation of ECMAScript.

It defines the rules and features of the language such as variables, functions, objects, operators, classes and control statements. ECMAScript is maintained through the ECMA International standardization process.

In simple words, ECMAScript provides the standard on which modern JavaScript is based.

Q10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead zone?

Ans:

Keyword	Scope	Reassignment	Hoisting
`var`	Function	Allowed	Hoisted and initialized as `undefined`
`let`	Block	Allowed	Hoisted but in TDZ
`const`	Block	Not allowed	Hoisted but in TDZ

Temporal Dead Zone (TDZ):

The TDZ is the time between entering a block and the declaration of a `let` or `const` variable. During this period, the variable exists but cannot be accessed before its declaration.

```
console.log(a); // Error
let a = 10;
```

Therefore, `let` and `const` are safer for block-scoped variables, while var has function scope.